



Fullstack Web Development

Session 1 | Week 1/Day 1 | 120 min

HTML5 - Part01



Instructor: Ms. Liev Nova

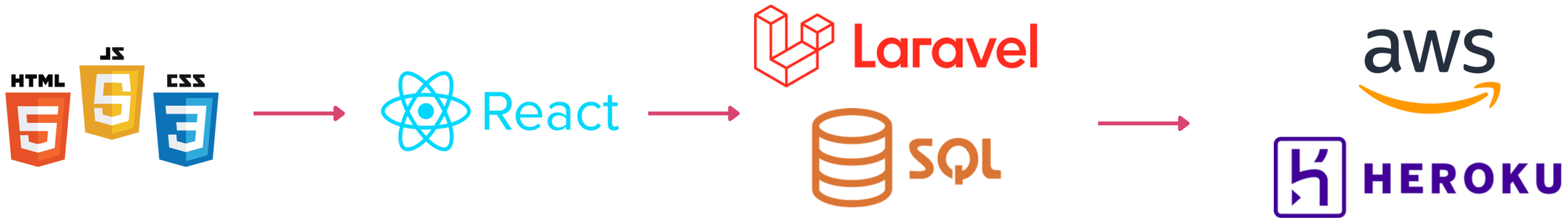


090667393

Introduction

12 weeks → 4 projects

(static site → fullstack app deployed to cloud)



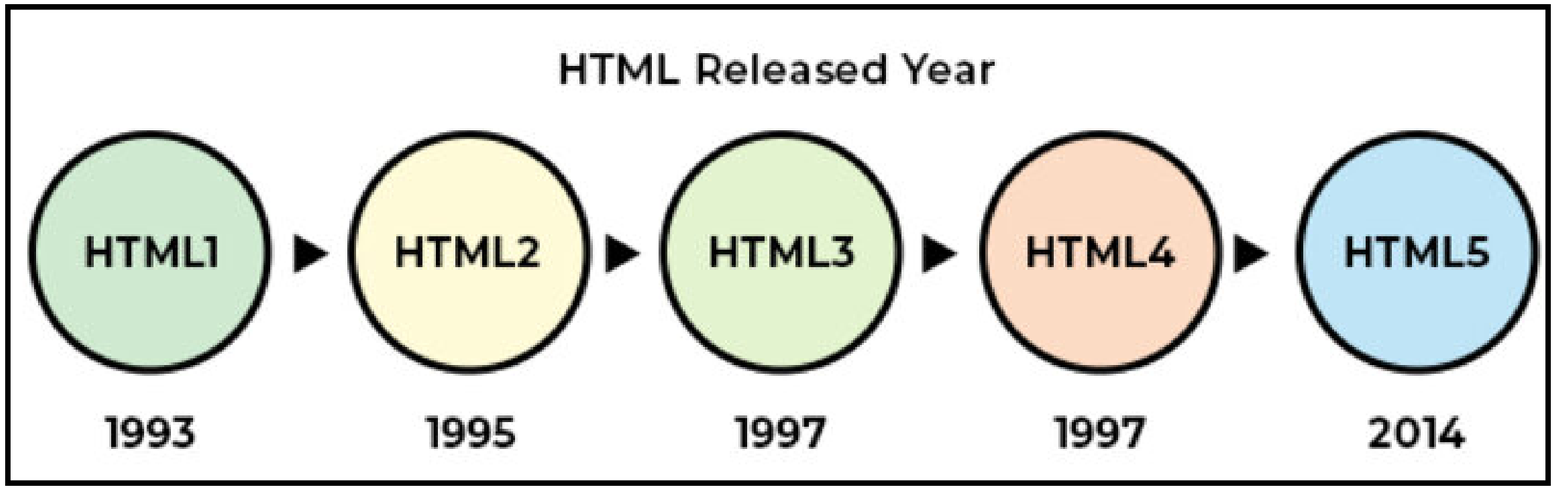
What is HTML ?

- **HTML** stands for **HyperText Markup Language**. It is used to design web pages using a markup language.
- **HTML** is the combination of **Hypertext** and **Markup language.**
- **Hypertext** defines the link between web pages.
- **A markup language** is used to define the text document within the **tag** which defines the structure of web pages.
- File extensions: **.html** or **.hml**

What is HTML ?

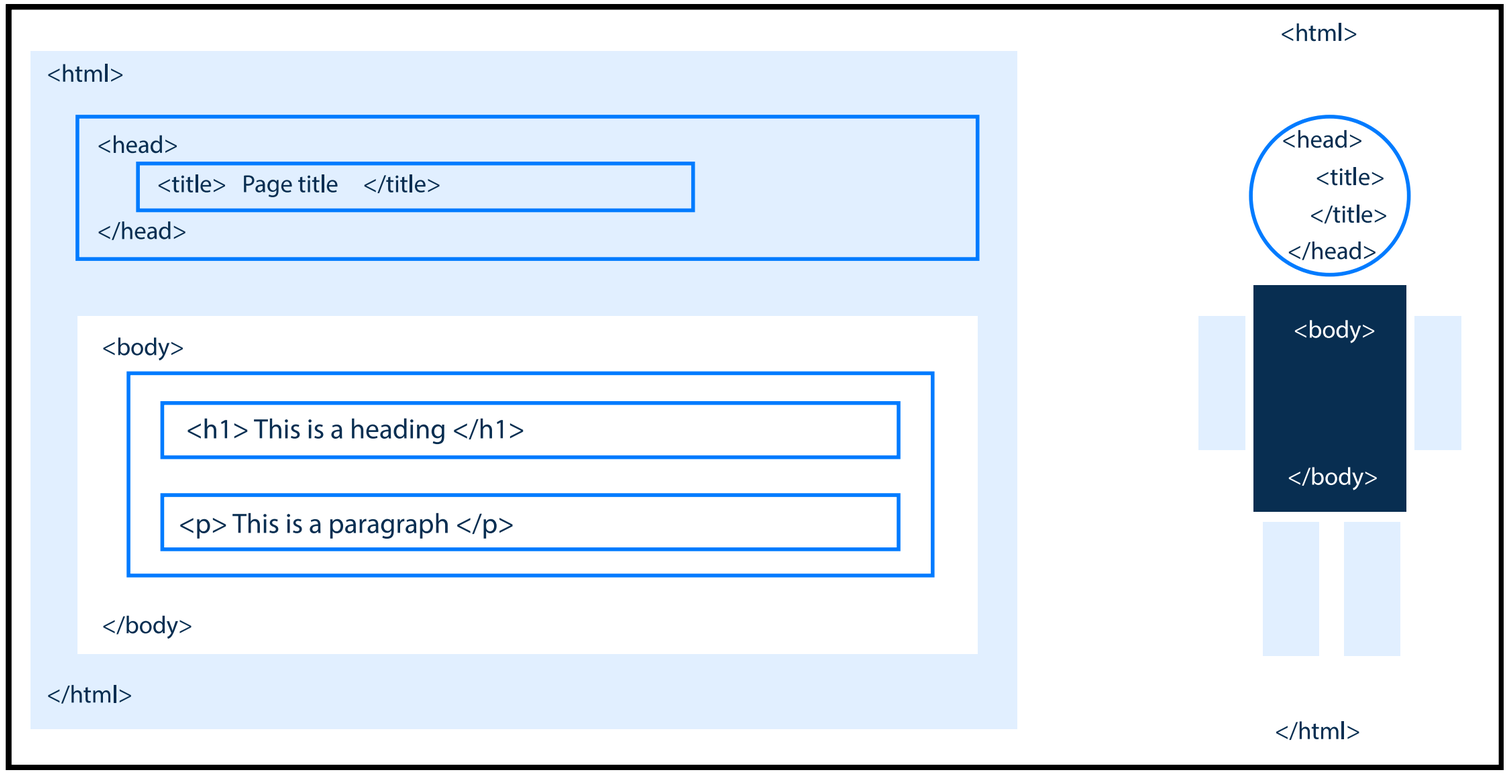
HTML Version History

- The first version of **HTML** was written by **Tim Berners-Lee** in **1993**



What is HTML ?

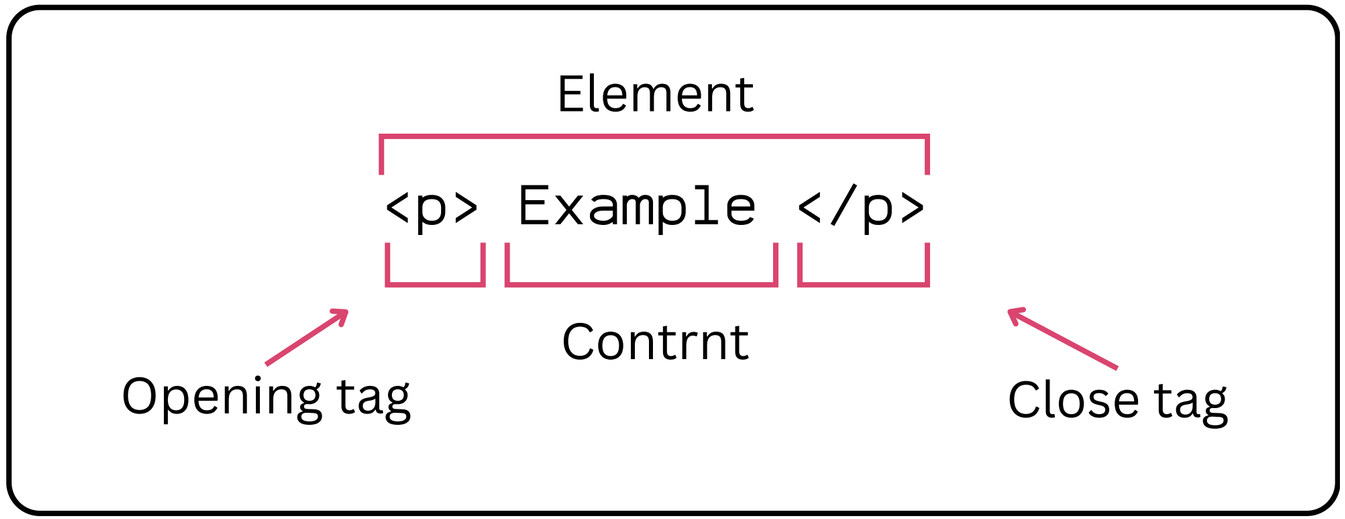
HTML Page Structure



What is HTML ?

HTML Tag

An HTML tag is a piece of markup language used to indicate the beginning and end of an HTML element in an HTML document.



What is HTML ?

HTML Attribute

- An **HTML attribute** is a piece of markup language used to adjust the behavior or display of an **HTML element**.
- For example, attributes can be used to change the color, size, or functionality of HTML elements.
- Attributes are used by including them in an opening HTML tag
- Different tag has different attribute

Syntax:

```
1   <tag_name attribute_name="value">Content</tag_name>
```

Example:

```
<tag_name style="font-size:40px">This text content will be sized to 40 pixels by the web browser.</tag_name>
```

What is HTML ?

Basic Structure of HTML Page

```
<! DOCTYPE html>
< html lang="en">
    < head>
        < meta charset="UTF-8">
        < meta http-equiv="X-UA-Compatible" content="IE=edge">
        < meta name="viewport" content="width=device-width, initial- scale=1.0">
        < title> Title of the page </ title>
    </ head>
    < body>
    </ body>
</ html>
```

What is HTML ?

The doctype

- At the start of each document, we have HTML5 Doctype declaration

```
1   <! DOCTYPE html>
```

- We can also use lowercase

```
1   <! doctype html>
```

What is HTML ?

The html tag and lang attribute

- The root tag for whole document is the head tag and you can find this tag after the Doctype declaration

```
1   < html lang="en">
2       < head>
```

- We usually have “en” that stands for English
- For the full list of supported languages, you can have a look at:

http://www.iana.org/assignments/language-subtag-registry/language-subtag-registry

What is HTML ?

The <META> tag

The <meta> tag is used to send information about the web page to search engines. Its most common attributes are name and content, which are used for name-value pairs

```
<meta name="description" content="A basic HTML boilerplate file.">
```

What is HTML ?

The <META> tag

- A list of common values for the name attribute are:
  - ***author***: Defines the name of the author of the document
  - ***description***: Provides the description of the document that should show up in search engine results
  - ***color-scheme***: Determines whether a page supports dark mode on devices that allow it.
  - ***viewport***: Gives information about the initial size of the document. This is used on mobile devices only.
  - ***robots***: Determines whether the document should be included in search engine results.

What is HTML ?

The html tag and lang attribute

- We also need to specify the character encoding, which in simple words will tell the browser how to parse the information in this document.
- The meta tag is a void element or self-closing HTML element which doesn’t require a closing tag

```
< meta charset="UTF-8">
```

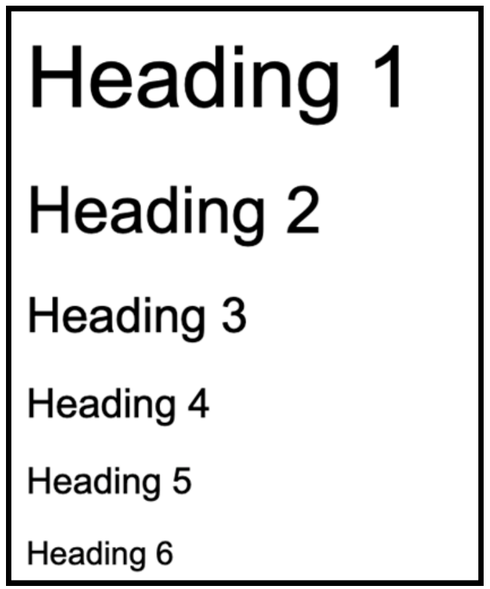
- The mostly used charset encoding is utf-8 there are also others if you need to specify them

What is HTML ?

Heading

- HTML headings are defined with the <h1> to <h6> tags.
- <h1> defines the most important heading. <h6> defines the least important heading

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

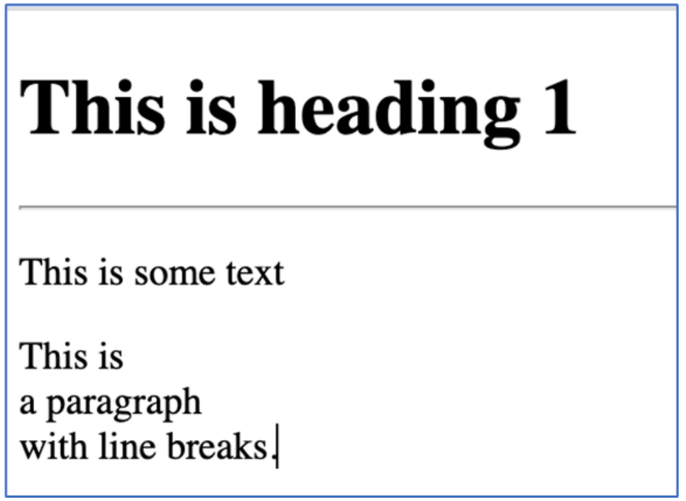


What is HTML ?

Paragraphs

Tag	Description
<p>	Defines a paragraph
<hr>	Defines a thematic change in the content
 	Inserts a single line break

```
<h1>This is heading 1</h1>
<hr>
<p>This is some text</p>
<p>This is<br>a paragraph<br>with line breaks.</p>
```



What is HTML ?

Formatting Elements

- Formatting elements were designed to display special types of text

Tag	Description
<b>	Defines bold text
	Defines emphasized text
<i>	Defines a part of text in an alternate voice or mood
<small>	Defines smaller text
	Defines important text
<sub>	Defines subscripted text
<sup>	Defines superscripted text
<ins>	Defines inserted text
<del>	Defines deleted text
<mark>	Defines marked/highlighted text

What is HTML ?

Formatting Elements

- Example

```
<p><b>This text is bold.</b></p>
<p><strong>This text is important!</strong></p>
<p><i>This text is italic</i></p>
<em>This text is emphasized</em>
<p>Do not forget to buy <mark>milk</mark> today.</p>
<small>This is some smaller text.</small>
<p>My favorite color is <del>blue</del> red.</p>
<p>My favorite color is <del>blue</del> <ins>red</ins>.</p>
<p>This is <sub>subscripted</sub> text.</p>
<p>This is <sup>superscripted</sup> text.</p>
```

This text is bold.

This text is important!

This text is italic

This text is emphasized

Do not forget to buy milk today.

This is some smaller text.

My favorite color is ~~blue~~ red.

My favorite color is ~~blue~~ red.

This is $_{\text{subscripted}}$ text.

This is ^{superscripted} text.

What is HTML ?

Quotation and Citation Elements

Tag	Description
<abbr>	Defines an abbreviation or acronym
<address>	Defines contact information for the author/owner of a document
<bdo>	Defines the text direction
<blockquote>	Defines a section that is quoted from another source
<cite>	Defines the title of a work
<q>	Defines a short inline quotation

What is HTML ?

HTML Lists

- Unordered HTML List

```
<ul>
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
</ul>
```

An unordered HTML list:

- Item
- Item
- Item
- Item

What is HTML ?

HTML Lists

- Ordered HTML List

```
<ol>
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
</ol>
```

An ordered HTML list:

1. First item
2. Second item
3. Third item
4. Fourth item

What is HTML ?

HTML Tables

- HTML tables allow web developers to arrange data into rows and columns.
- Example:

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico

What is HTML ?

HTML Tables

- Table tags

Tag	Description
<table>	Defines a table
<th>	Defines a header cell in a table
<tr>	Defines a row in a table
<td>	Defines a cell in a table
<caption>	Defines a table caption
<colgroup>	Specifies a group of one or more columns in a table for formatting
<col>	Specifies column properties for each column within a <colgroup> element
<thead>	Groups the header content in a table
<tbody>	Groups the body content in a table
<tfoot>	Groups the footer content in a table

What is HTML ?

HTML Tables

- Table Attributes

  - border: table border
  - colspan: merge columns
  - rowspan: merge row

- Table Style

  - border-collapse : border style
  - padding: cell padding
  - text-align: cell content alignment
  - nth-child: select odd or even row

What is HTML ?

HTML Tables

- Code Table Example

```
<body>
    <h2>A basic HTML table</h2>
    <table style="width:100%">
        <tr>
            <th>Company</th>
            <th>Contact</th>
            <th>Country</th>
        </tr>
        <tr>
            <td>Alfreds Futterkiste</td>
            <td>Maria Anders</td>
            <td>Germany</td>
        </tr>
        <tr>
            <td>Centro comercial Moctezuma</td>
            <td>Francisco Chang</td>
            <td>Mexico</td>
        </tr>
    </table>
    <p>To understand the example better, we have added borders to the table.</p>
</body>
```

A basic HTML table

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico

To understand the example better, we have added borders to the table.

What is HTML ?

HTML Links

- HTML links are hyperlinks
- You can click on a link and jump to another document
- Note: A link does not have to be text. A link can be an image or any other HTML element!
- HTML Links – Syntax:
  - The HTML <a> tag defines a hyperlink. It has the following syntax:

```
<a href="url">link text</a>
```

What is HTML ?

HTML Links

- When you move the mouse over a link, the mouse arrow will turn into a little hand.
- By default, links will appear as follows in all browsers:
  - An unvisited link is underlined and blue
  - A visited link is underlined and purple
  - An active link is underlined and red

What is HTML ?

The target Attribute

- By default, the linked page will be displayed in the current browser window. To change this, you must specify another target for the link.
- The target attribute specifies where to open the linked document.
- The target attribute can have one of the following values:
  - _self - Default. Opens the document in the same window/tab as it was clicked
  - _blank - Opens the document in a new window or tab
  - _parent - Opens the document in the parent frame
  - _top - Opens the document in the full body of the window

What is HTML ?

Absolute URLs vs. Relative URLs

- A local link (a link to a page within the same website) is specified with a relative URL (without the "https://www" part):
- An absolute link is a hyperlink containing a full URL to a specific webpage or document.

What is HTML ?

Absolute URLs vs. Relative URLs

```
<h2>Absolute URLs</h2>
<p><a href="https: //www.w3.org/">W3C</a></p>
<p><a href="https: //www.google.com/">Google</a></p>
```

```
<h2>Relative URLs</h2>
<p><a href="html_images.asp">HTML Images</a></p>
<p><a href=" /css/default.asp">CSS Tutorial</a></p>
```

What is HTML ?

HTML Link Target Types

- **In page**: a link to specific part of document within the same page
  - Define link: ...
  - Define document to link to: <tag id="TARGET">...</tag>
- **Internal**: a link to specific page within the same domain. A link target could exclude the domain
  - Link text

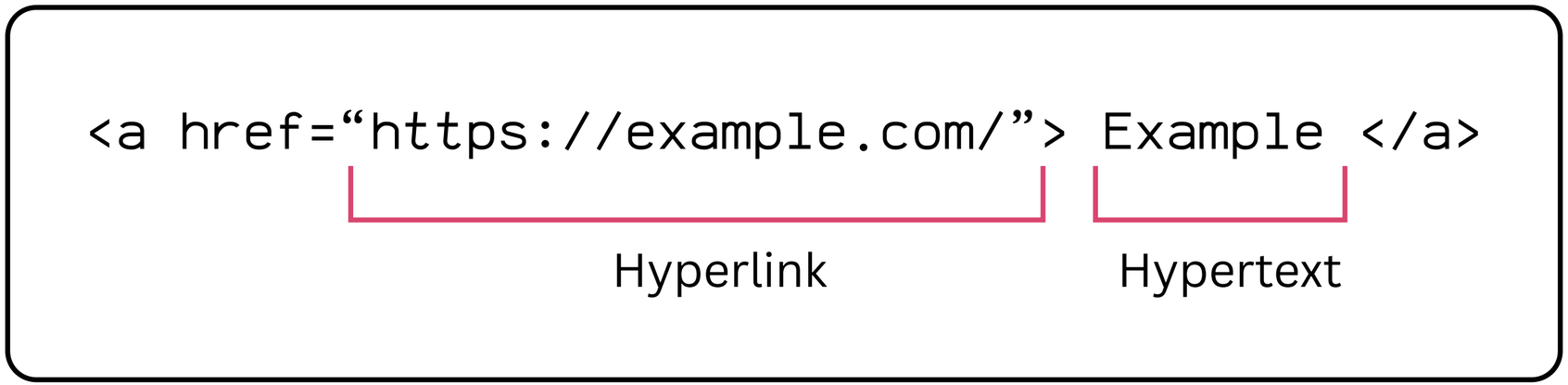
What is HTML ?

HTML Link Target Types

- **External**: a link to external page
  - Google
- **Special link**: to email client or phone
  - Email: liev.nova@gmail.com
  - Phone: +85590667393

What is HTML ?

Hyperlink vs Hypertext



What is HTML ?

HTML IFrame

- An HTML iframe is used to display a web page within a web page.

Syntax

```
<iframe src="url" title="description"></iframe>
```

- Always include a title attribute (for screen readers)
- The height and width attributes specify the size of the iframe
- Use border:none to remove the border around the iframe
- Use style to apply CSS style.

What is HTML ?

HTML <picture> Element

Display image base on the view port

```
<picture>
    <source media="(min-width:650px)" srcset="img_pink_flowers.jpg">
    <source media="(min-width:465px)" srcset="img_white_flower.jpg">
    <img src="img_orange_flowers.jpg" alt="Flowers" style="width:auto;"
</picture>
```

What is HTML ?

HTML <audio> Tag

- The <audio> tag is used to embed sound content in a document, such as music or other audio streams.
- The <audio> tag contains one or more <source> tags with different audio sources. The browser will choose the first source it supports.

```
<audio controls  >
    <source src="horse.ogg" type="audio /ogg">
    <source src="horse.mp3" type="audio /mpeg">
    Your browser does not support the audio tag.
</audio>
```

What is HTML ?

Audio Format and Browser Support

Browser	MP3	WAV	OGG
Edge / IE	YES	YES*	YES*
Chrome	YES	YES	YES
Firefox	YES	YES	YES
Safari	YES	YES	NO
Opera	YES	YES	YES

*From Edge 79

What is HTML ?

HTML <video> Tag

- The <video> tag is used to embed video content in a document, such as a movie clip or other video streams.
- The <video> tag contains one or more <source> tags with different video sources. The browser will choose the first source it supports.

```
<video width="320" height="240" controls>
    <source src="movie.mp4" type="video /mp4">
    <source src="movie.ogg" type="video /ogg">
    Your browser does not support the video tag.
</video>
```

What is HTML ?

Video Format and Browser Support

Browser	MP4	WebM	Ogg
Edge	YES	YES	YES
Chrome	YES	YES	YES
Firefox	YES	YES	YES
Safari	YES	YES	NO
Opera	YES	YES	YES

HTML5 Semantics Theory

HTML5 = Modern Web Foundation

```
<div class="header-area">
    <div class="logo-container">
        <div class="nav-wrapper">
            <div class="menu-item">Home</div>
        </div>
    </div>
</div>
```

Non-Semantic HTML

```
<header>
    <nav>
        <ul>
            <li><a href="#home">Home</a></li>
        </ul>
    </nav>
</header>
```

Semantic HTML

HTML5 Semantics Theory

7 Semantics tages you must memorize

Tag	Purpose	Real Example	**Never Use For**
<header>	Page banner + logo/nav	Top of portfolio	Footer content
<nav>	**Main navigation only**	4-6 menu links	Footer links
<main>	**Primary content**	Hero/About/Portfolio	Sidebars
<section>	Themed content block	"About Me" section	Single paragraph
<article>	Standalone content	Blog post, testimonial	Page layout
<aside>	Secondary content	Sidebar, social links	Main content
<footer>	Page ending	Copyright + social	Headers

HTML5 Form

Form field you must memorize

Type	Use Case	Validation
text	Name, subject	maxlength, minlength
email	Contact forms	Auto @ validation
tel	Phone numbers	pattern="\+855[0-9]{8}"
number	Age, quantity	min, max
date	Birthdays, events	Calendar picker
select	Dropdowns	<option> lists
textarea	Long text	rows, maxlength



THANK YOU

For your attention !

